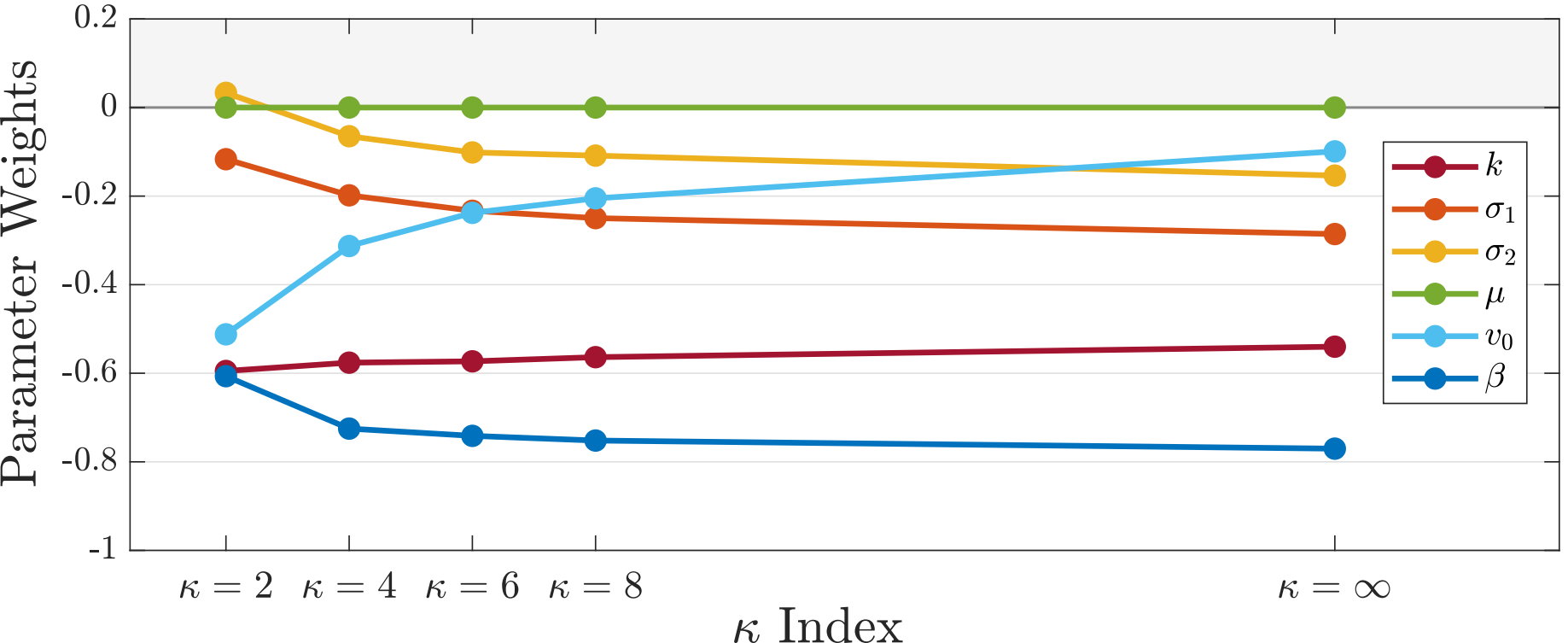Active Weight Vector of Bump-on-Tail Distributions